

IEOR 4404: Simulation

Introduction and Probability Review

Henry Lam

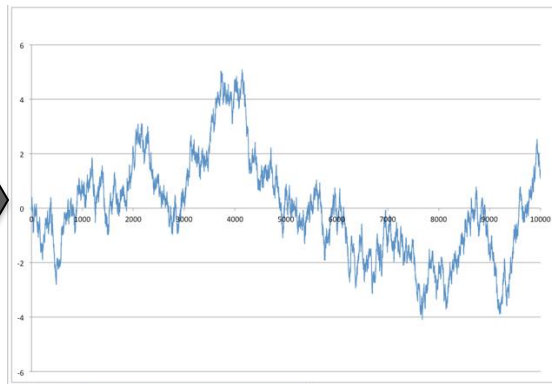
Overview

Why simulation?

- Models are used to describe real-world phenomena
- Rarely are these models analytically tractable



Real world



Stochastic model

Theoretical analysis

$$E[\sup_{0 \leq s \leq t} B_s] = (2t/\pi)^{0.5}$$



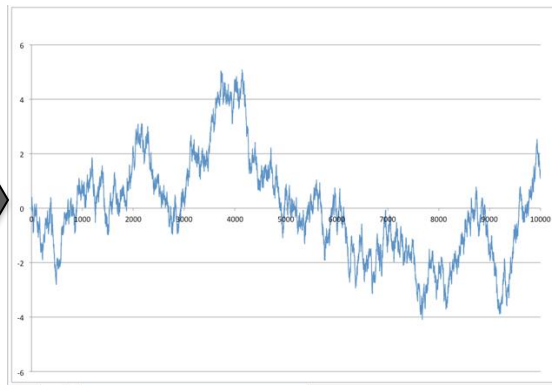
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Simulation

Motivation

Simulation is a computation tool to:

- Estimate quantities such as performance measures from a model
- Assist decision making

Motivation

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- Managing a flight with 100 seats – revenue of \$200 each
- May be no-shows

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- If overcapacity, penalty of \$300 for each overcapacitated seat

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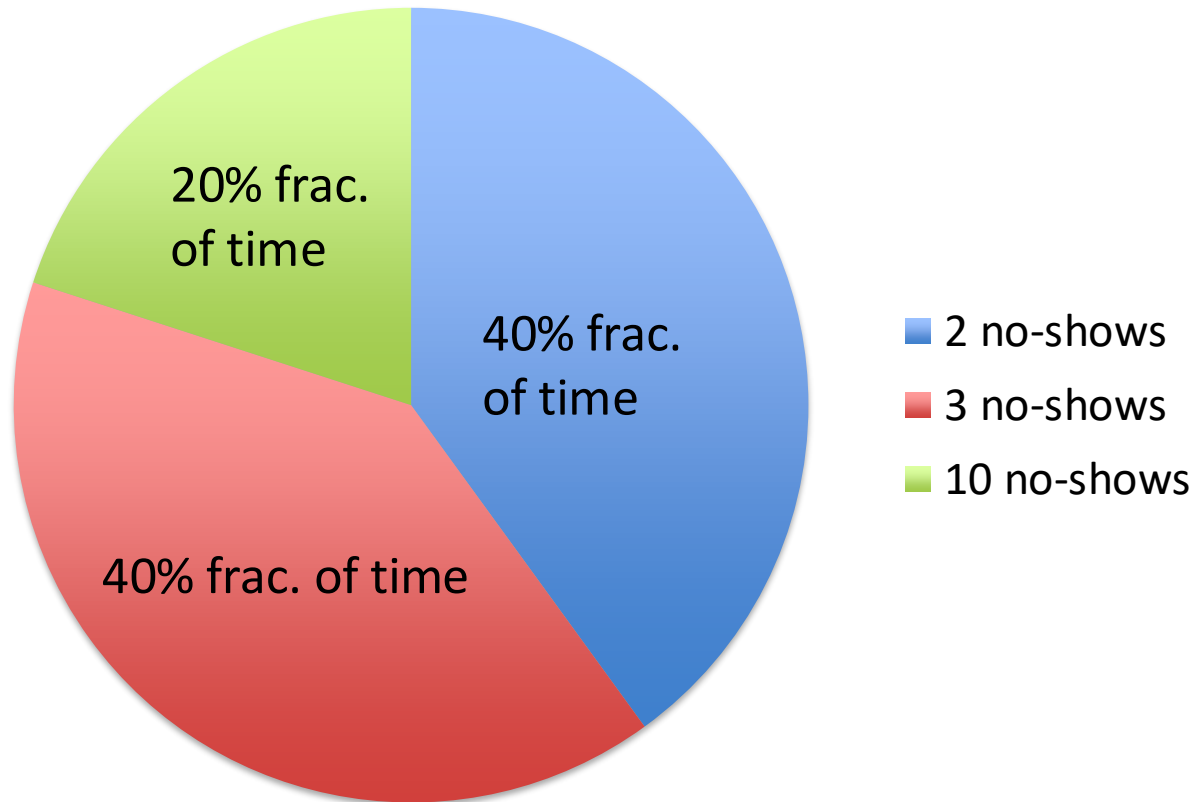
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-
1. Figure out the number of no-shows D .
 - A random quantity
 - Understand this randomness based on **historical data**.

Motivation

Suppose the following distribution is available, obtained via statistical analysis on data

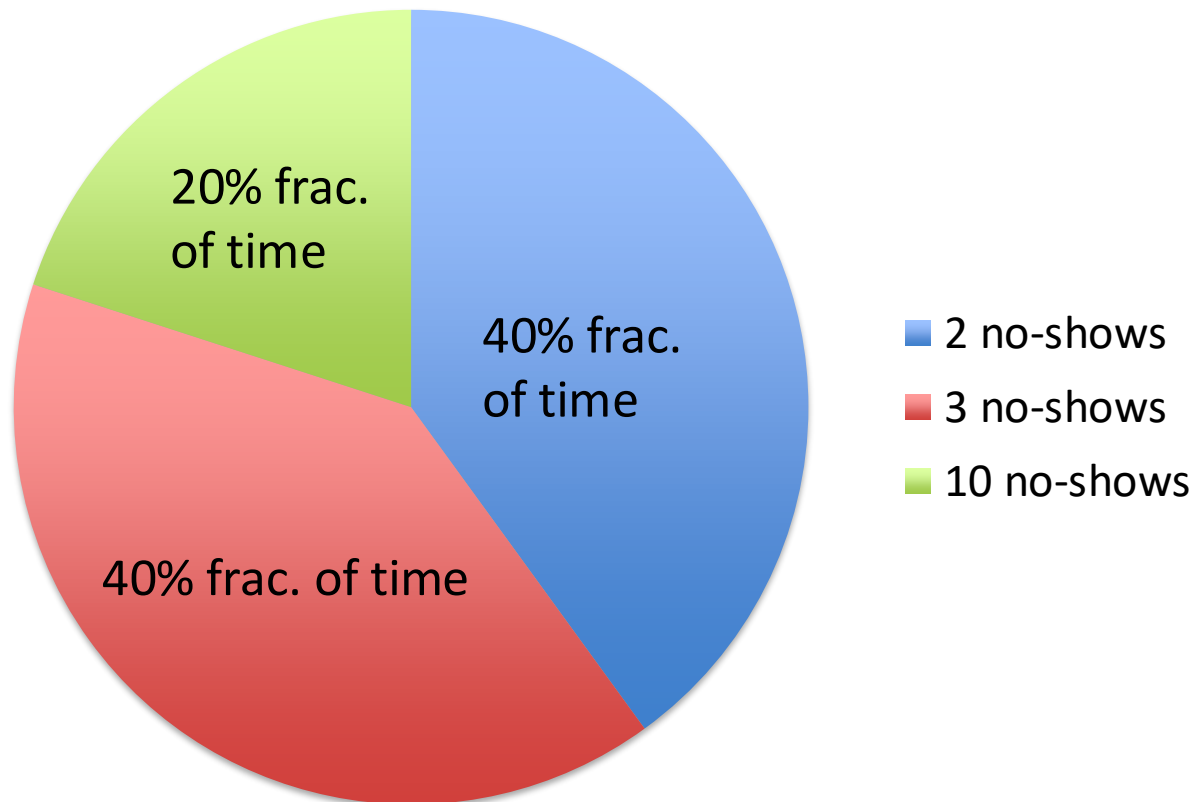


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1. Figure out the number of no-shows D .
 2. Next step?

Motivation

- “Average” # of no-shows = $2 \times 0.4 + 3 \times 0.4 + 10 \times 0.2 = 4$.
- Intuitively, oversell 4 seats?



Motivation

- “Average” # of no-shows = $2 \times 0.4 + 3 \times 0.4 + 10 \times 0.2 = 4$.
- If I oversell 4 seats (an intuitive solution):
 - If 2 no-shows, then revenue = $\$200 \times 104 - \$300 \times 2 = \$20,200$
 - If 3 no-shows, revenue = $\$200 \times 104 - \$300 = \$20,500$
 - If 10 no-shows, then revenue = $\$200 \times 104 = \$20,800$
- Total *average* revenue = $\$20,200 \times 0.4 + \$20,500 \times 0.4 + \$20,800 \times 0.2 = \$20,440$.

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- Oversell 3 seats? *Average* revenue = $\$20480 > \20440 .
- Oversell 2 seats? *Average* revenue = $\$20400$

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- Oversell 2 seats? *Average* revenue = \$20400
- Optimal is to oversell 3 seats

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- This is a **decision-making** problem, where the decision is to determine the number of over-sells
- This is a **stochastic** problem, where the number of no-shows is random
- In order to make the optimal decision, we need to **estimate** the average revenue for different numbers of over-sells

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-
1. Figure out the number of no-shows D .
 2. Analyze the decision-making problem.
 3. Often times the problem is much more complicated than this example – resort to **simulation**

Motivation

A more complicated problem can consist of:

- Distribution D can say range from 1 to 10 (10 possible realizations)
- Need to distribute among 10 different flights.
 - Altogether, $10 \times 10 \times \dots \times 10 = 10^{10}$ possible realizations
- Interested in minimizing $P(\text{revenue} < 20400)$
- Have a limit for no. of tickets to oversell
- Need to make decisions on a multistage time horizon

As soon as problem gets more complicated, need simulation.

Topics

- Probability review
 - Probability space, conditional probability and independence
 - Random variables (discrete and continuous), expectation, variance, covariance, correlation, common distributions
- Generating random variables
 - Often used to generate the inputs that are fed into a system logic to transform into target outputs
- Output analysis
 - Statistical analysis on the outputs
- Simulating stochastic processes or systems
 - Translating inputs into complex outputs
- Variance reduction
 - Speed up simulation

Foundation of Probability

- Start with a real-world situation or a *random experiment*
- Set of *all* possible outcomes is the *sample space*
- ***Probabilities*** are assignments of the likelihoods on these possible outcomes

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 - A single die roll. Sample space is {1, 2, ..., 6}. A fair die assigns probability $\frac{1}{6}$ to each outcome.
 - Repeatedly throw a fair die until the first six shows up. Sample space is {1, 2, ...}, where k indicates that the first six shows up on the kth roll. Assign probabilities $\frac{1}{6}$, $(\frac{5}{6}) \times (\frac{1}{6})$, $(\frac{5}{6})^2 \times (\frac{1}{6})$, ... to outcome 1, 2, 3, ...

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- Examples of events:
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 - The event that “an even number shows in a die roll” is the subset $\{2, 4, 6\}$
 - 3 consecutive coin tosses: what is the subset for the event “no Tail occurs in the 2nd toss”?

$\left\{ \begin{array}{l} H H H, \\ H H T, \\ T H H, \\ T H T \end{array} \right\}$

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- An ***event*** is *a set* of outcomes in the sample space
- Examples of events:
 - The event that “an even number shows in a die roll” is the subset $\{2, 4, 6\}$
 - 3 consecutive coin tosses: what is the subset for the event “no Tail occurs in the 2nd toss”?
- One can assign a probability to any event using rules of probability

Specifying Probabilities

- A sample space contains n outcomes. If I pick one outcome in a fully random manner, the probability that I pick outcome i is

$$P(\text{selecting outcome } i) = \frac{1}{n}$$

- Event A consists of k outcomes. If I pick one outcome in a fully random manner, the probability that event A happens is

$$P(\text{event } A) = \frac{k}{n}$$

Counting

- To understand further how to specify probabilities, we also need to know how to count.
- There are k containers. If container i has n_i objects, and all objects are distinct. I pick one object from each container. The total number of possible outcomes is

$$n_1 \times n_2 \times \cdots \times n_k$$

Counting

- A permutation of n distinct objects is an ordering of these objects
- A permutation of k objects out of n distinct objects is an ordering of any k objects out of the n ones
- The total number of permutations of n distinct objects is
$$n! = n \times (n - 1) \times (n - 2) \times \cdots \times 1$$
- The total number of permutations of k distinct objects out of n ones is

$${}_nP_k = n \times (n - 1) \times (n - 2) \times \cdots \times (n - k + 1) = \frac{n!}{(n - k)!}$$

Counting

- A combination of k objects out of n distinct objects is a collection of any k objects out of the n ones (ordering does not matter)
- The total number of combinations of k distinct objects out of n ones is

$$\binom{n}{k} = \frac{n!}{k! (n - k)!}$$

Algebra of Events

- We often want to form “composite” events from simpler events
- Example: pick a random card from a deck of 52 cards.
 - A = event that the chosen card is an ace
 - B = event that the chosen card is a heart
 - May be interested in the event that the card is an ace or a heart – $A \cup B$ (read “ A union B ”)
 - Or the event that the card is *both* an ace and a heart – $A \cap B$ (read “ A intersection B ”)

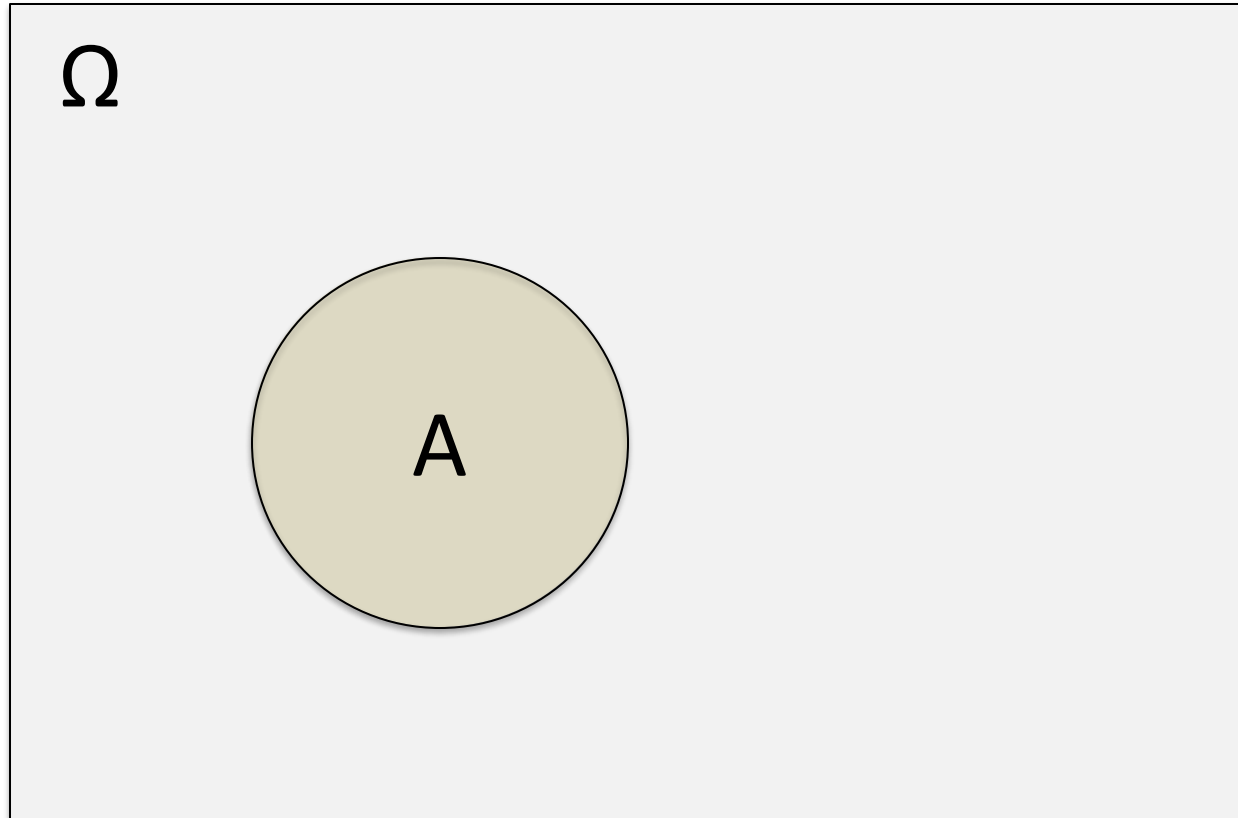
Algebra of Events

- $A \cup B$ is the event that A or B occurs

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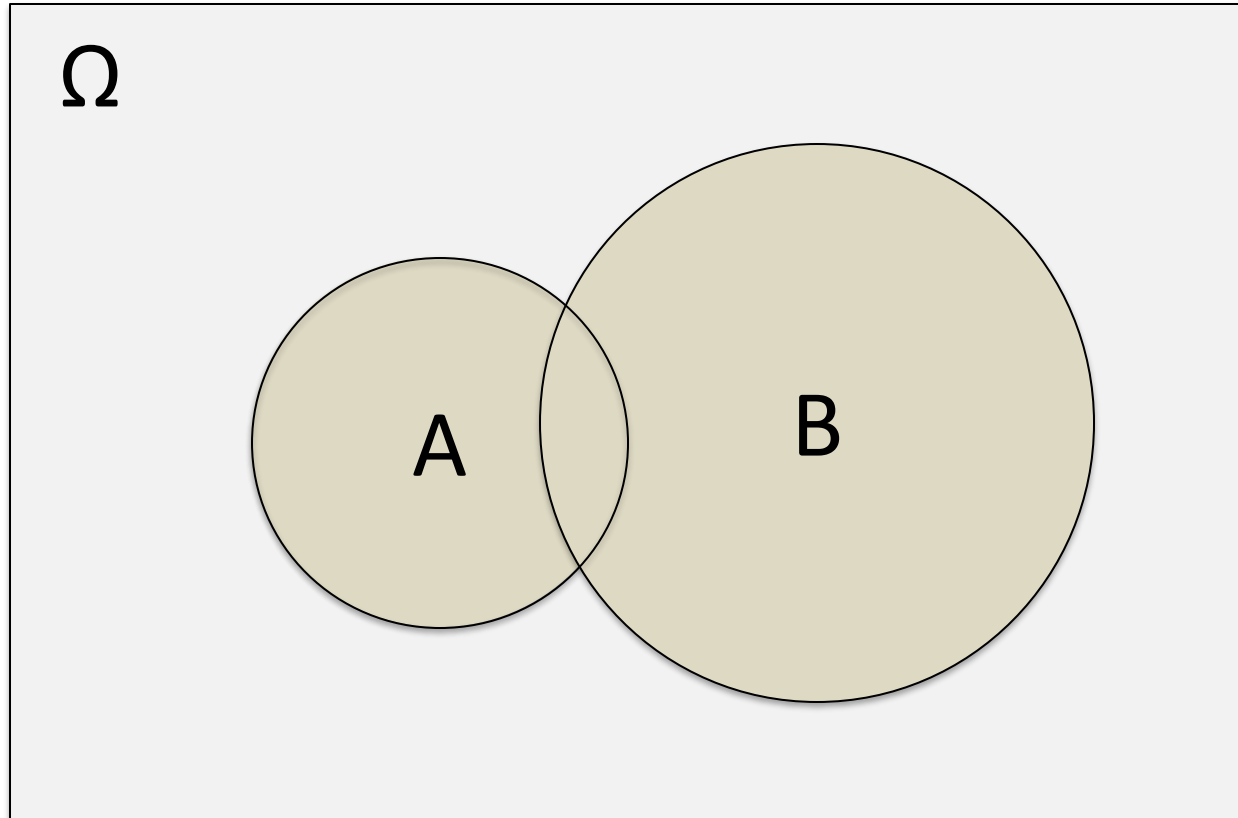
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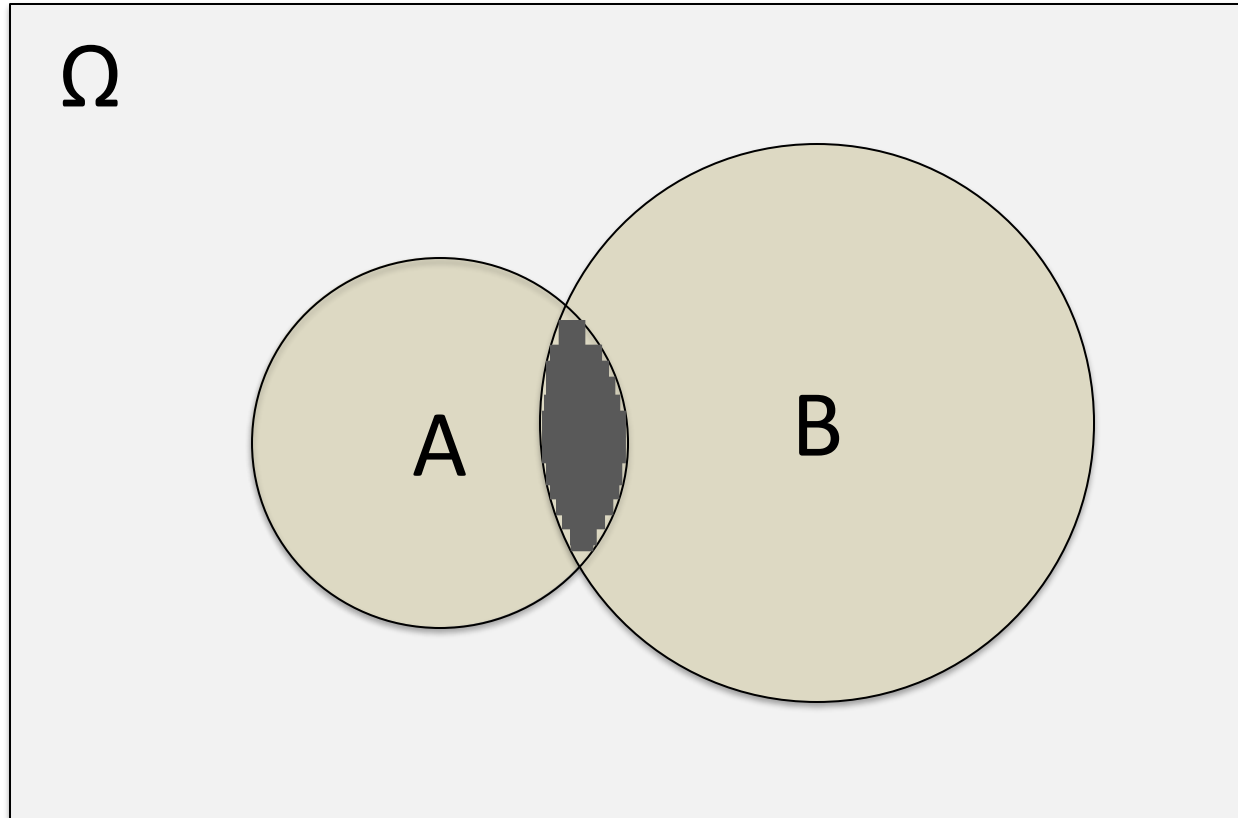
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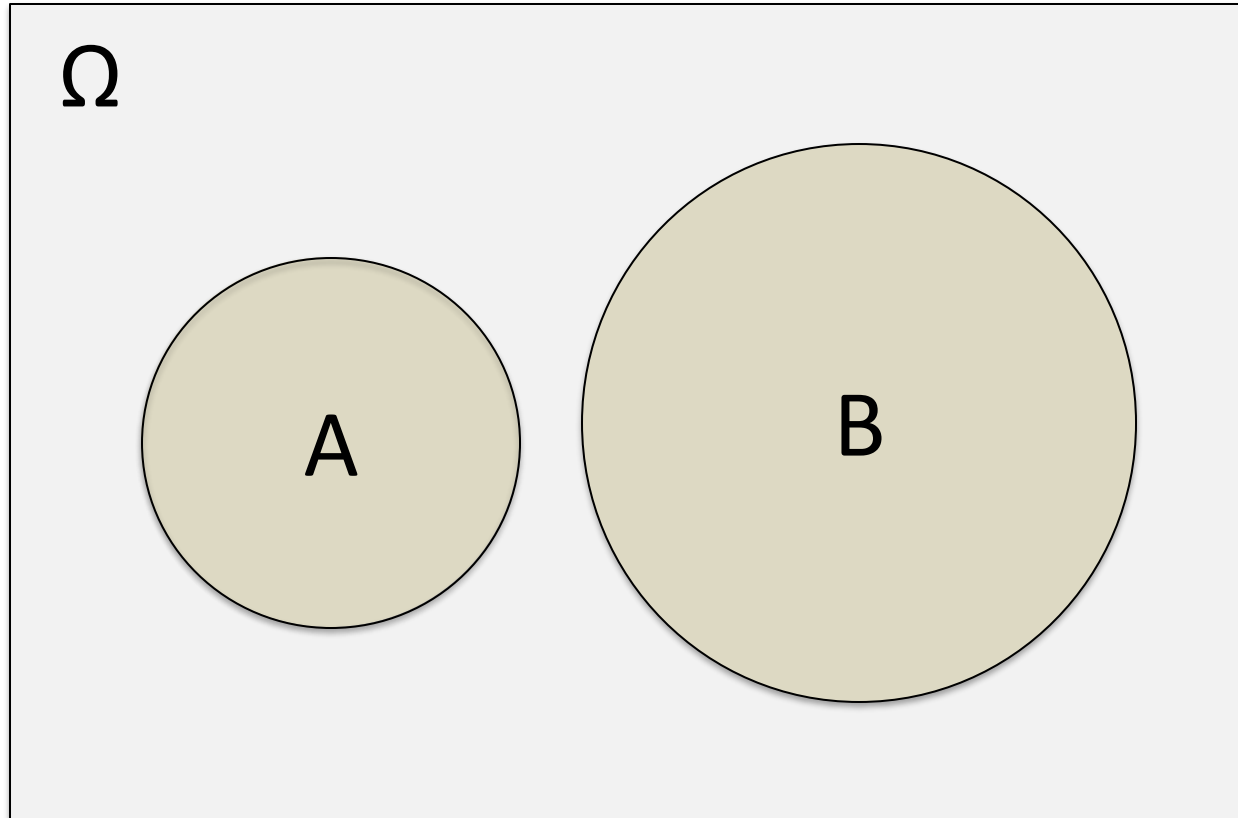
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Algebra of Events

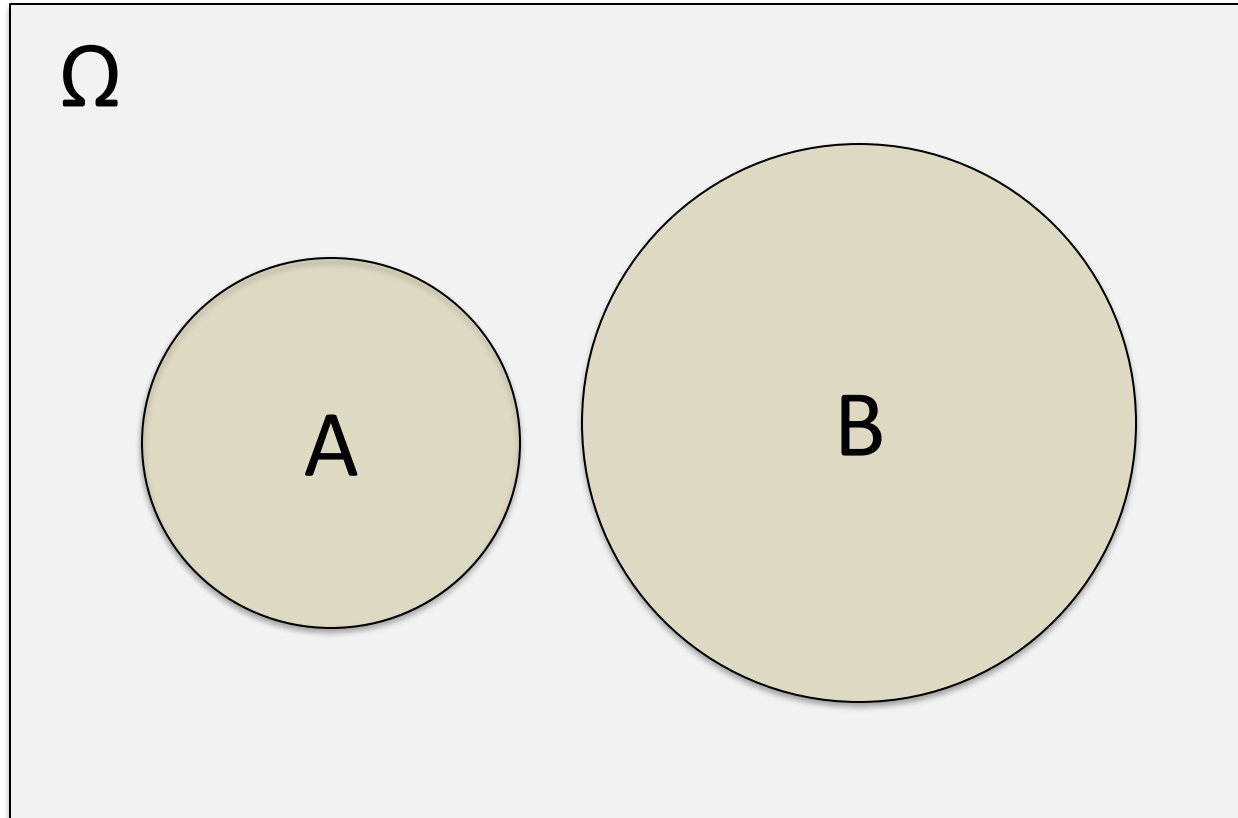
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- In this case, A and B are called disjoint or mutually exclusive

Algebra of Events

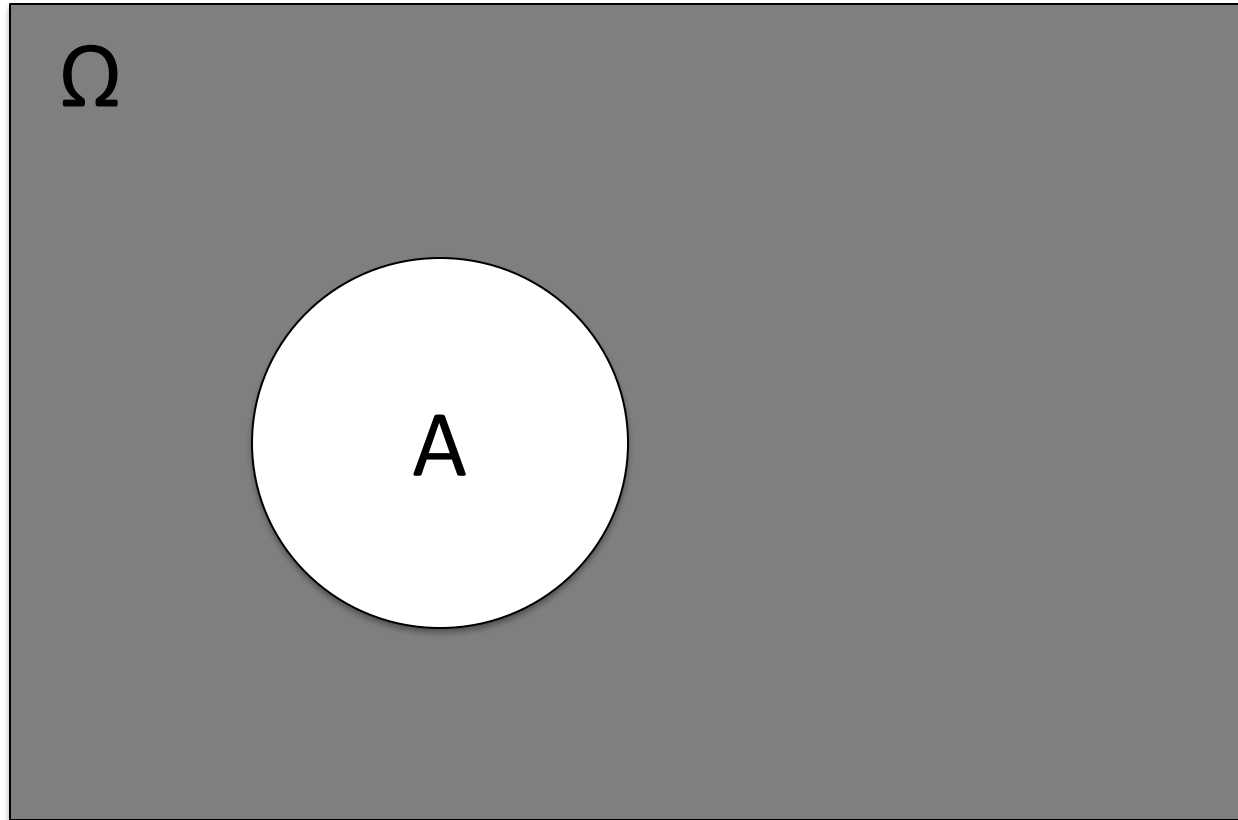
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- In this case, A and B are called disjoint or mutually exclusive
- Q: is the set $\{0\}$ empty?

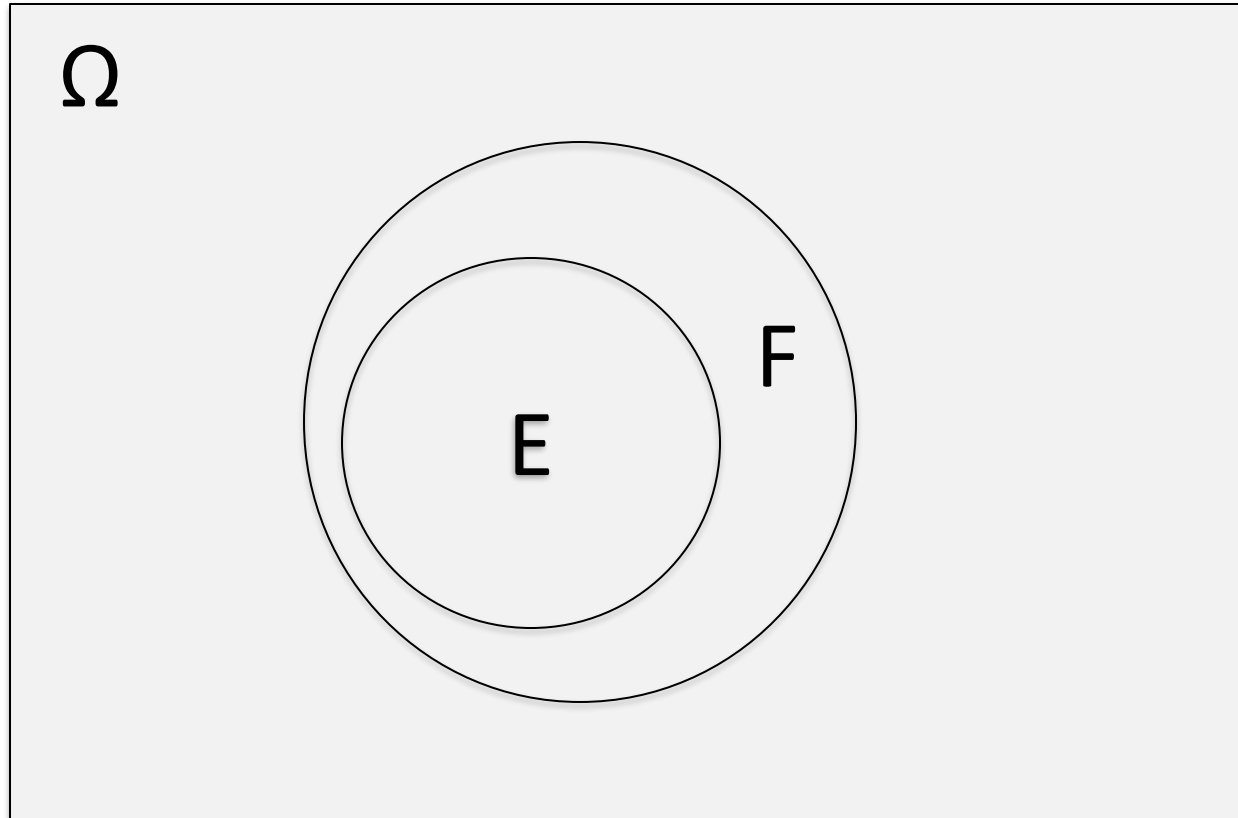
Algebra of Events

- A^c is the complement of A – shaded area



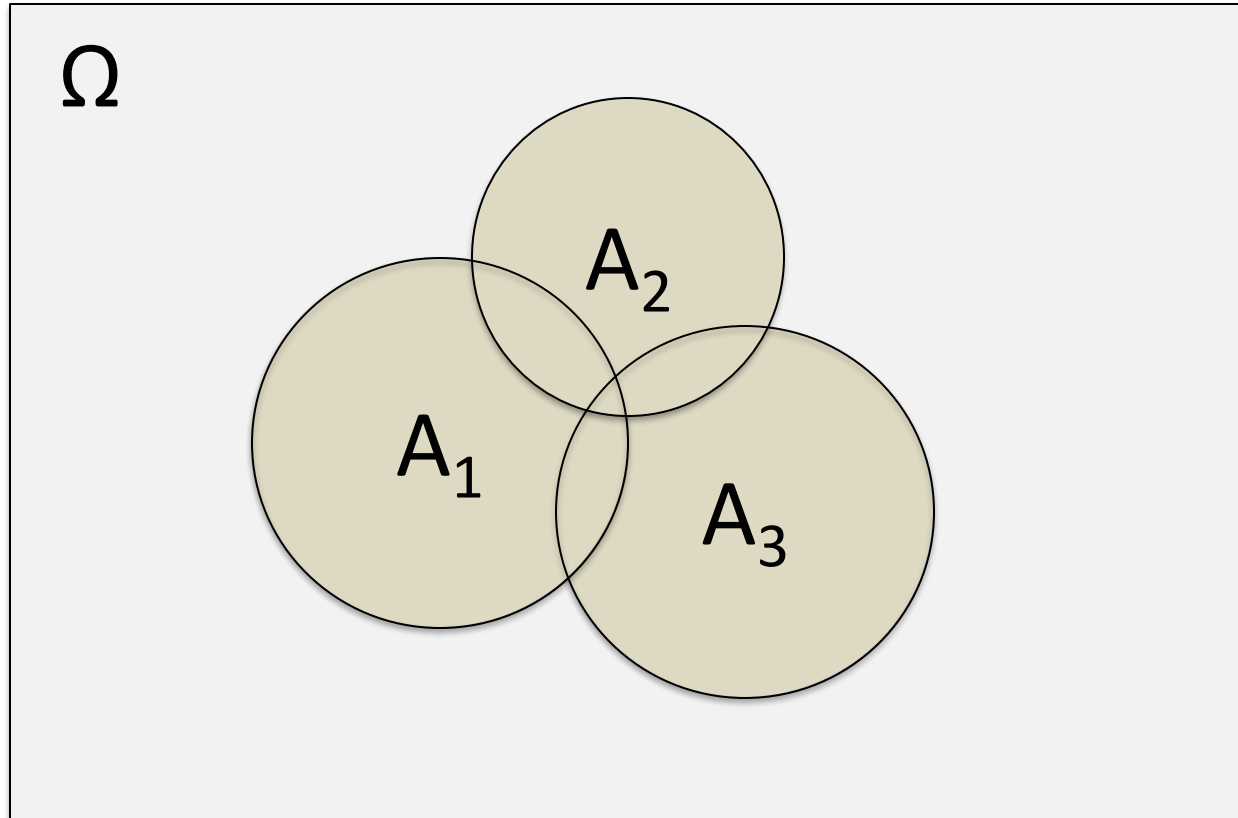
Algebra of Events

- $E \subseteq F$: E is a *subset* of F



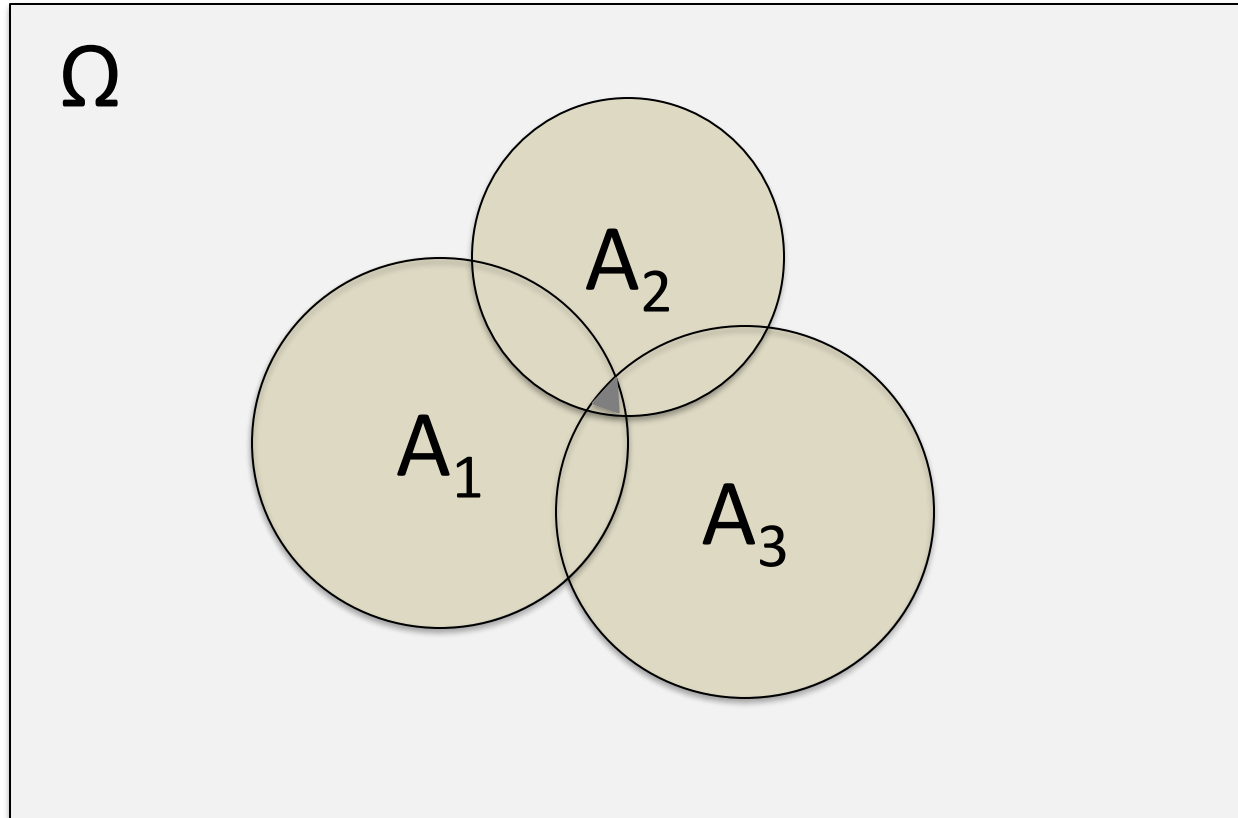
Algebra of Events

- Can also form $A_1 \cup A_2 \cup \dots \cup A_n$, the union of more than 2 events



Algebra of Events

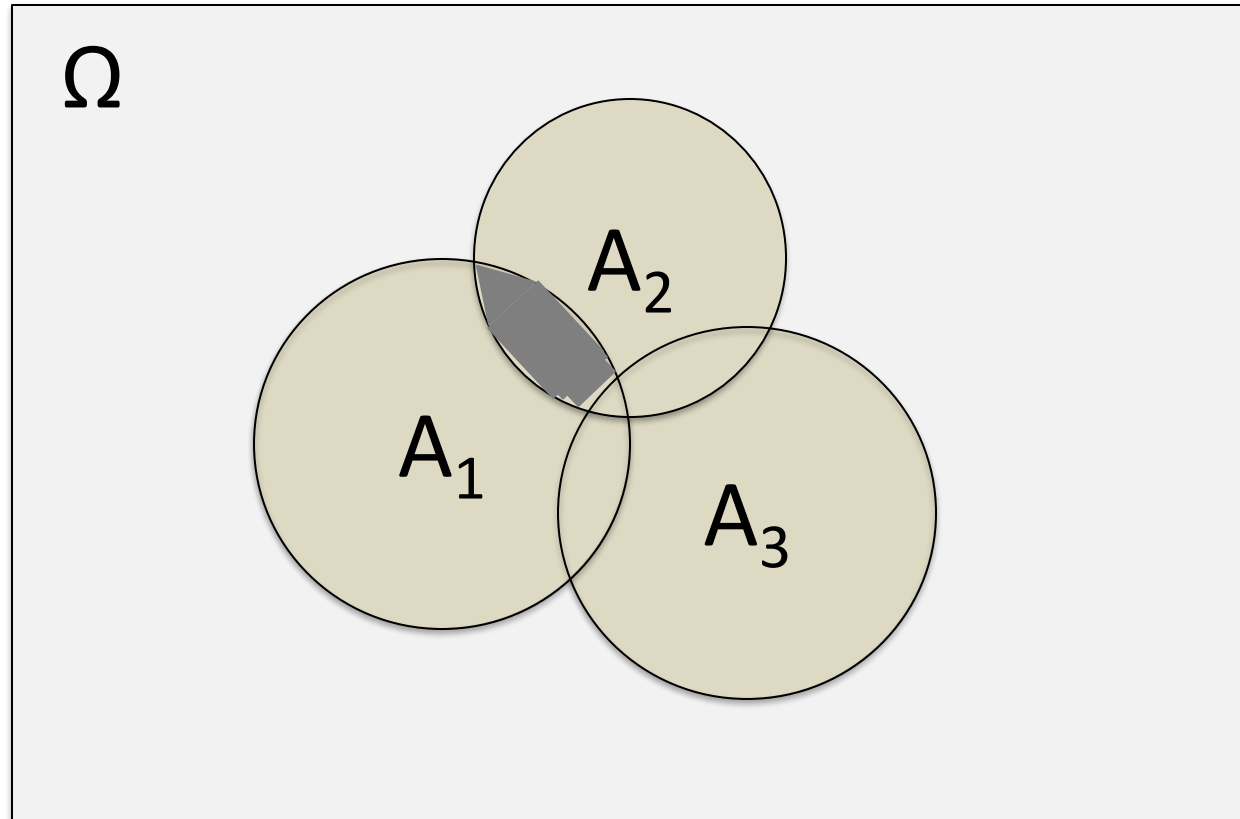
- Can also form $A_1 \cup A_2 \cup \dots \cup A_n$, the union of more than 2 events



- Similarly, can form $A_1 \cap A_2 \cap \dots \cap A_n$, the intersection of more than 2 events

Algebra of Events

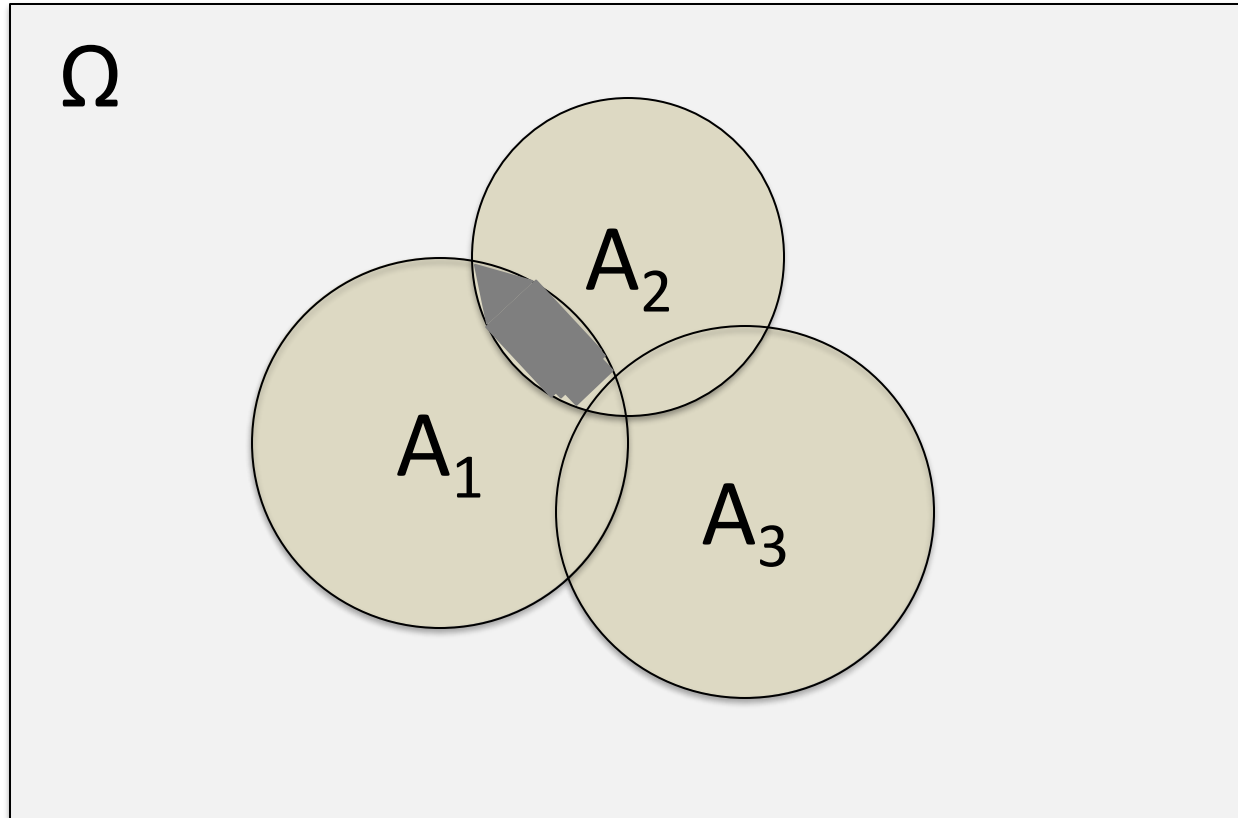
- Q: What is the shaded area in terms of A_1 , A_2 , A_3 , $\cap$, $\cup$, and complement?



$$A_1 \cap A_2 \cap A_3^c$$

Algebra of Events

- Q: What is the shaded area in terms of A_1 , A_2 , A_3 , $\cap$, $\cup$, and complement?



- Ans: $A_1 \cap A_2 \cap A_3^c$

Axioms of Probability

- *Probabilities, events, and sample space* are three pillars of a probability model
- Probabilities are assigned to events of a sample space
- The assignment need to satisfy some properties:
 - **Axiom 1:** $0 \leq P(A) \leq 1$
 - **Axiom 2:** $P(\Omega) = 1$
 - **Axiom 3:** Let $A_1, A_2, \dots$ be mutually exclusive events. Then $P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$

Axioms of Probability

- Further properties that we can derive
- **Property 1:** $P(A^c) = 1 - P(A)$
 - Ex. A = “it rains tomorrow,” and $P(A) = 30\%$; then $P(A^c)$, the prob. that it *doesn't* rain tomorrow is $1 - 30\% = 70\%$.
- **Property 2:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$A \cup A^c = \Omega \quad \Rightarrow \quad P(A \cup A^c) = P(\Omega)$$

$$P(A \cup A^c) = P(A) + P(A^c) \quad \text{by Axiom 3}$$

because A & A^c
are mutually
exclusive

$$P(\Omega) = 1 \quad \text{by Axiom 2}$$

Axioms of Probability

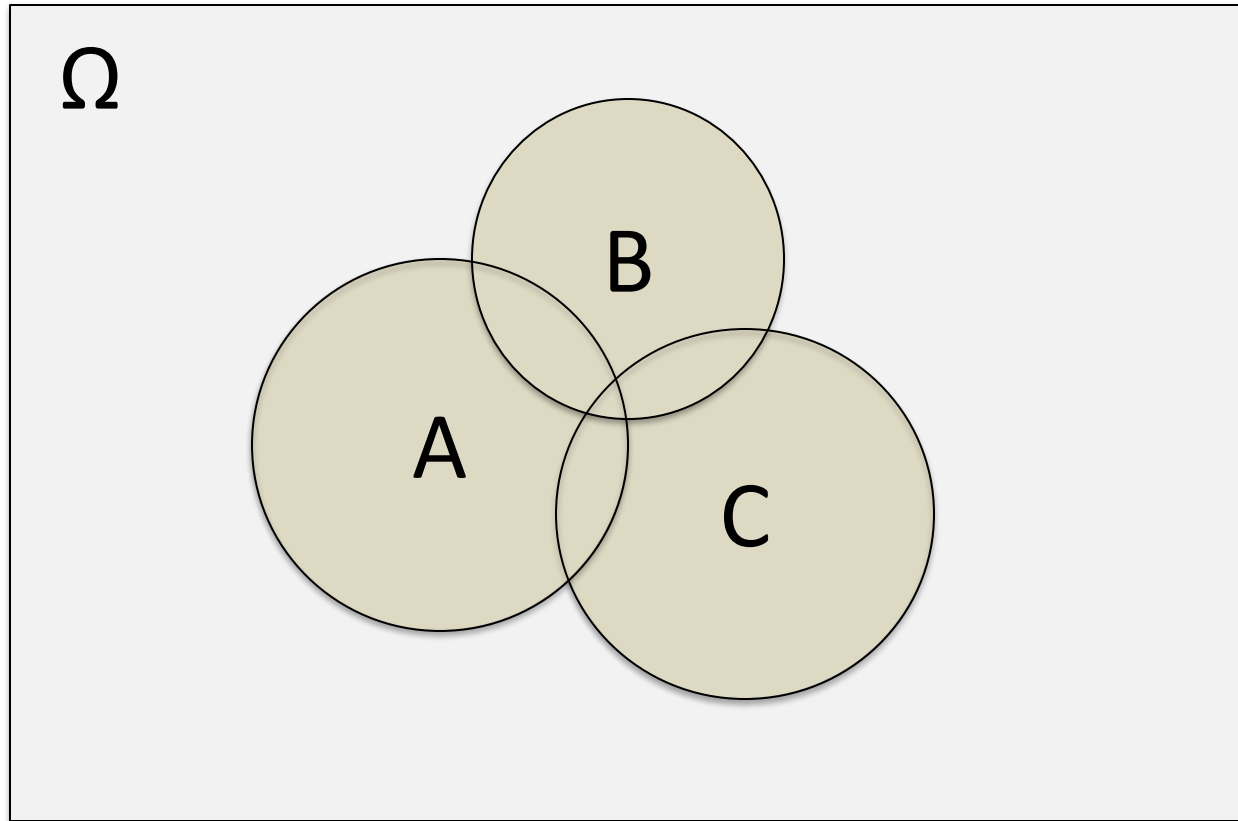
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 - What is the prob of a randomly drawn card being an ace or a heart?

Axioms of Probability

- Return to example
 - What is the prob of a randomly drawn card being an ace or a heart?
 - A = event that it's an ace
 - B = event that it's a heart
 - $P(A) = 1/13$
 - $P(B) = 1/4$
 - $P(A \cap B) = 1/52$
 - $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $= 1/13 + 1/4 - 1/52 = 16/52 = 4/13$

Inclusion-Exclusion Principle

- $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$



Conditional Probability

- **Definition:** For two events A and B (with $P(B) > 0$), the conditional probability $P(A | B)$ is defined to be
$$P(A | B) = P(A \cap B) / P(B)$$
- This is the probability of A given the knowledge that B occurs

Conditional Probability

- Example.** A bin contains 5 defective, 10 partially defective and 25 acceptable transistors. A transistor is picked at random and put to use. If it does not immediately fail, what is the probability that it is acceptable?

$$P(A|B)$$

A = acceptable

B = does not immediately fail

= partially defective or acceptable

$$= \frac{P(A \cap B)}{P(B)}$$

$$= \frac{P(A)}{P(B)}$$

$$= \frac{25 / 40}{35 / 40} = \frac{25}{35}$$

Conditional Probability

- **Example.** A bin contains 5 defective, 10 partially defective and 25 acceptable transistors. A transistor is picked at random and put to use. If it does not immediately fail, what is the probability that it is acceptable?
- **Solution.** $P(\text{acceptable} | \text{not defective})$
= $P(\text{acceptable and not defective}) / P(\text{not defective})$
= $P(\text{acceptable}) / P(\text{not defective})$
= $(25/40) / (35/40)$
= $5/7$

Conditional Probability

- **Example.** A family has two children. Given that there is at least one girl in the family, what's the probability that there are two girls?

$$A = 2 \text{ G}$$

$$B = \text{at least 1 G}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)}{P(B)} = \frac{1/4}{3/4} = \frac{1}{3}$$

$$\text{sample space: } \{ GG, GB, BG, BB \}$$

$$A = \{ GG \}$$

$$B = \{ GG, GB, BG \}$$

Conditional Probability

- **Example.** A family has two children. Given that there is at least one girl in the family, what's the probability that there are two girls?
- **Example.** A family has two children. Given that the older child is a girl, what's the probability that there are two girls?

$$\begin{aligned} P(A|B) & \quad B = \text{older is G} \\ &= \frac{1/4}{2/4} \\ &= \frac{1}{2} \end{aligned}$$

Conditional Probability

- Can rewrite the conditional probability formula as
$$P(A \cap B) = P(A|B)P(B)$$
- Sometimes called the **multiplication rule**
- Can expand the formula: $P(A \cap B \cap C) = P(C|A \cap B)P(B|A)P(A)$.

Conditional Probability

Example: You know that with 30% probability that your company will open a new branch in New York. You are also 60% certain that if the branch is opened you will become its new manager. What is the probability that you will be a new branch manager?

$$30\% \times 60\%$$

Independence

- **Definition:** two events A and B are independent if
$$P(A \cap B) = P(A)P(B)$$

- Recall the conditional probability formula

$$P(A \cap B) = P(A|B)P(B)$$

If $P(A) = P(A|B)$, i.e., occurrence of A does not depend on that of B, then A is independent of B

- Thus, an equivalent definition of independence between A and B is $P(A)=P(A|B)$, and also $P(B)=P(B|A)$
- Note: if $P(A) = P(A|B)$, then $P(A) = P(A|B^c)$ (can you prove it?)

$$P(A|B^c) = \frac{P(A \cap B^c)}{P(B^c)}$$

$\nwarrow$

$$P(A \cap B^c) = P(A) - P(A \cap B)$$

because $P(A \cap B) + P(A \cap B^c) = P(A)$
by Axioms 2 & 3.

$$P(A \cap B^c) = P(A) - P(A \cap B)$$

$$= P(A) - P(A)P(B)$$

because
A & B are indep

$$P(B^c) = 1 - P(B)$$

$$\frac{P(A \cap B^c)}{P(B^c)} = \frac{P(A) - P(A)P(B)}{1 - P(B)} = P(A)$$

Equivalent definitions of independence:

$$P(A \cap B) = P(A) P(B)$$

$$P(A) = P(A|B)$$

$$P(B) = P(B|A)$$

$$P(A) = P(A|B^c)$$

$$P(A^c) = P(A^c|B)$$

$$P(A^c) = P(A^c|B^c)$$

...

Independence

- **Example:** two fair dice are thrown. Let A be the event that the number shown by the first die is even, and B the event that the sum of the dice is odd. Are A and B independent?

$$P(A) = \frac{1}{2}$$

$$P(B) = \frac{1}{2}$$

$$P(A \cap B) = P(1^{st} = \text{even}, 2^{nd} = \text{odd})$$

$$= \frac{1}{4}$$

A & B are indep because $P(A \cap B) = P(A)P(B)$

Independence

- **Example:** two fair dice are thrown. Let A be the event that the number shown by the first die is even, and B the event that the sum of the dice is odd. Are A and B independent?
- **Answer:** they are independent
- $P(A) = 1/2$
 $P(B) = 1/2$ – (odd, even) or (even, odd) both result in an odd sum; each happens with probability $9/36 = 1/4$
 $P(A \cap B) = P(\text{first die is even and second die is odd}) = 1/4 = 1/2 * 1/2$

Independence

- What about independence of more than 2 events?
- For three events A, B and C: they are independent if $P(A \cap B \cap C) = P(A)P(B)P(C)$, $P(A \cap B) = P(A)P(B)$, $P(B \cap C) = P(B)P(C)$ and $P(A \cap C) = P(A)P(C)$
- For $A_1, A_2, \dots, A_n$, they are independent if for every subcollection $A_{1'}, A_{2'}, \dots, A_{r'}$, we have
$$P(A_{1'} \cap A_{2'} \cap \dots \cap A_{r'}) = P(A_{1'})P(A_{2'}) \dots P(A_{r'})$$

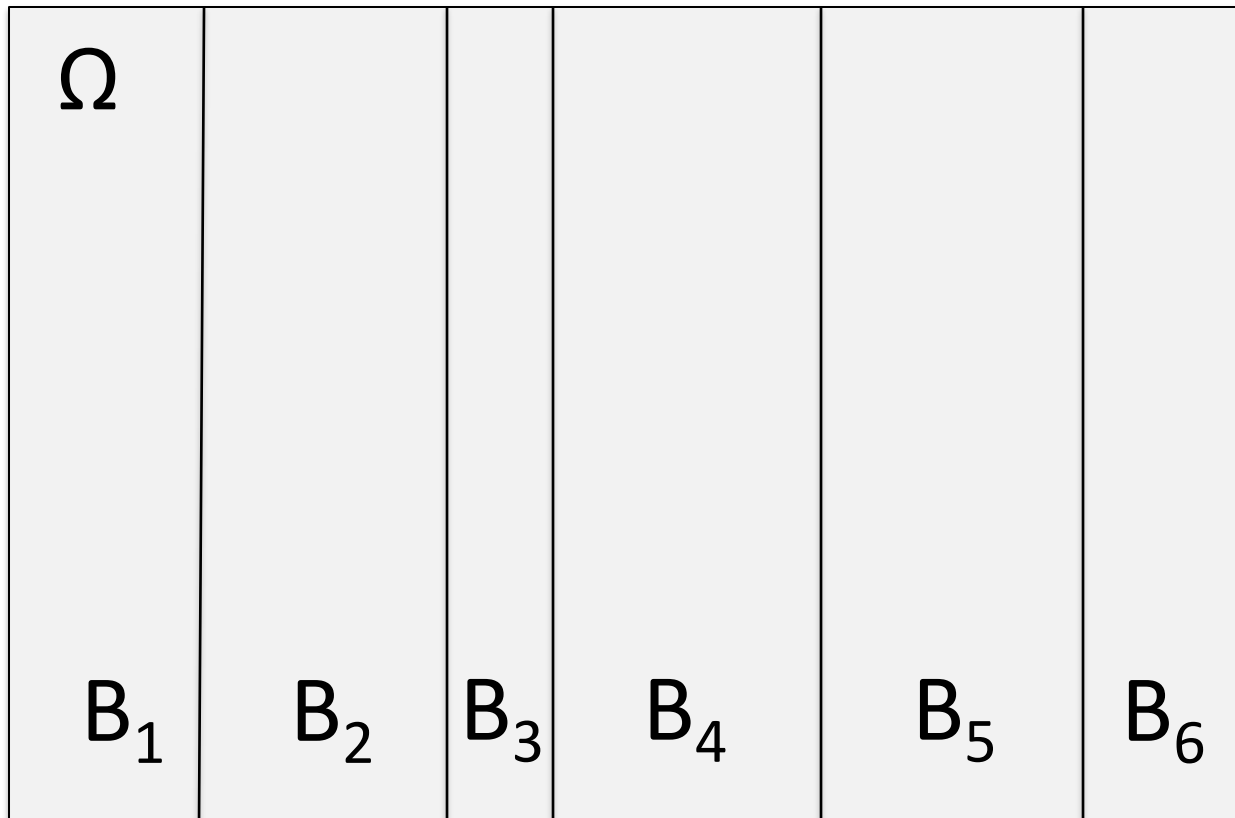
if only $P(A \cap B) = P(A)P(B)$, $\dots$, $P(A \cap C) = P(A)P(C)$,

then it's pairwise independent.

Generally, pairwise independence $\neq$ independence.

Law of Total Probability

- Let $B_1, B_2, \dots, B_n$ be a collection of mutually exclusive events, so that whenever $i \neq j$, $B_i \cap B_j = \emptyset$
- Furthermore, suppose that $B_1 \cup B_2 \cup \dots \cup B_n = \Omega$ is the whole sample space
- $B_1, B_2, \dots, B_n$ said to be a *partition* of Ω



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- $B_1, B_2, \dots, B_n$ said to be a *partition* of Ω
- Then for any event A ,

$$P(A) = P(A \cap B_1) + P(A \cap B_2) + \dots + P(A \cap B_n);$$

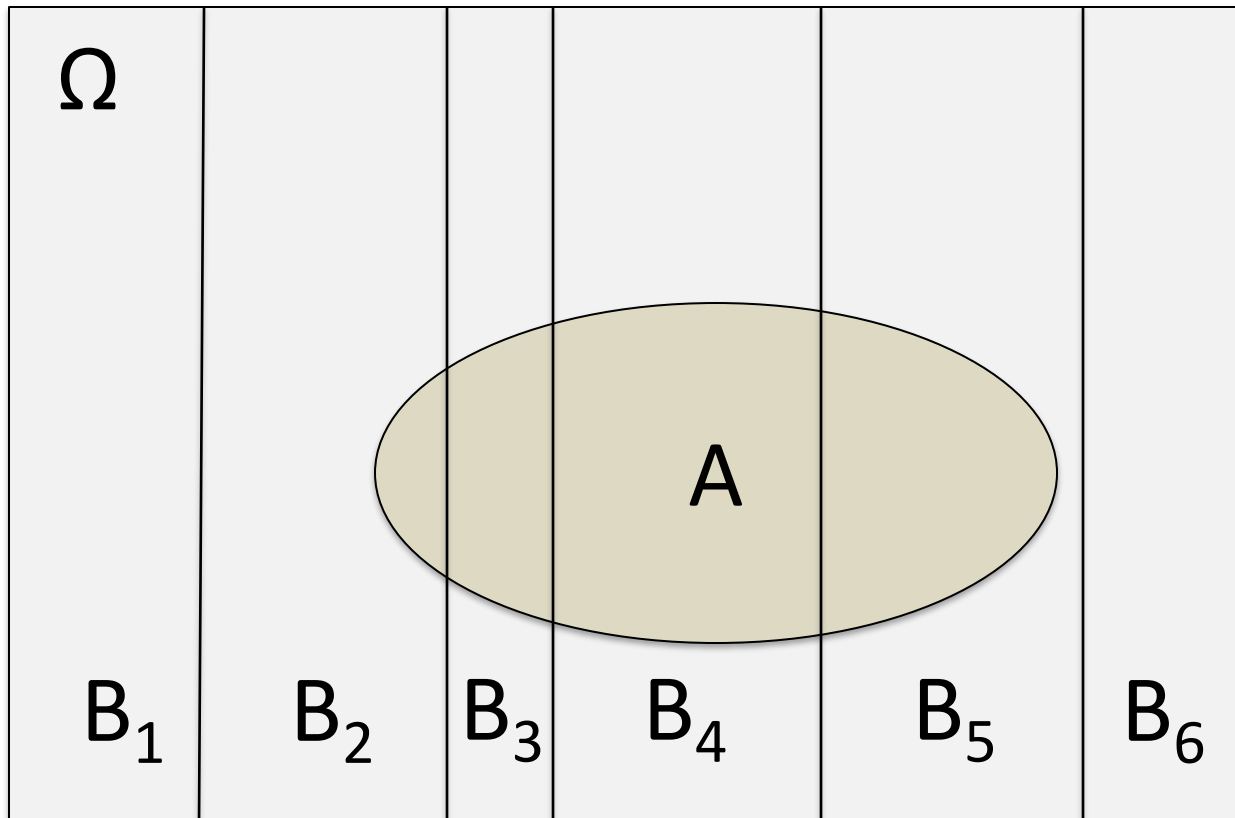
$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots + P(A|B_n)P(B_n)$$

- Can extend to infinitely many $B_1, B_2, \dots$

$$A = (A \cap B_1) \cup (A \cap B_2) \cup \dots \cup (A \cap B_n)$$

Law of Total Probability

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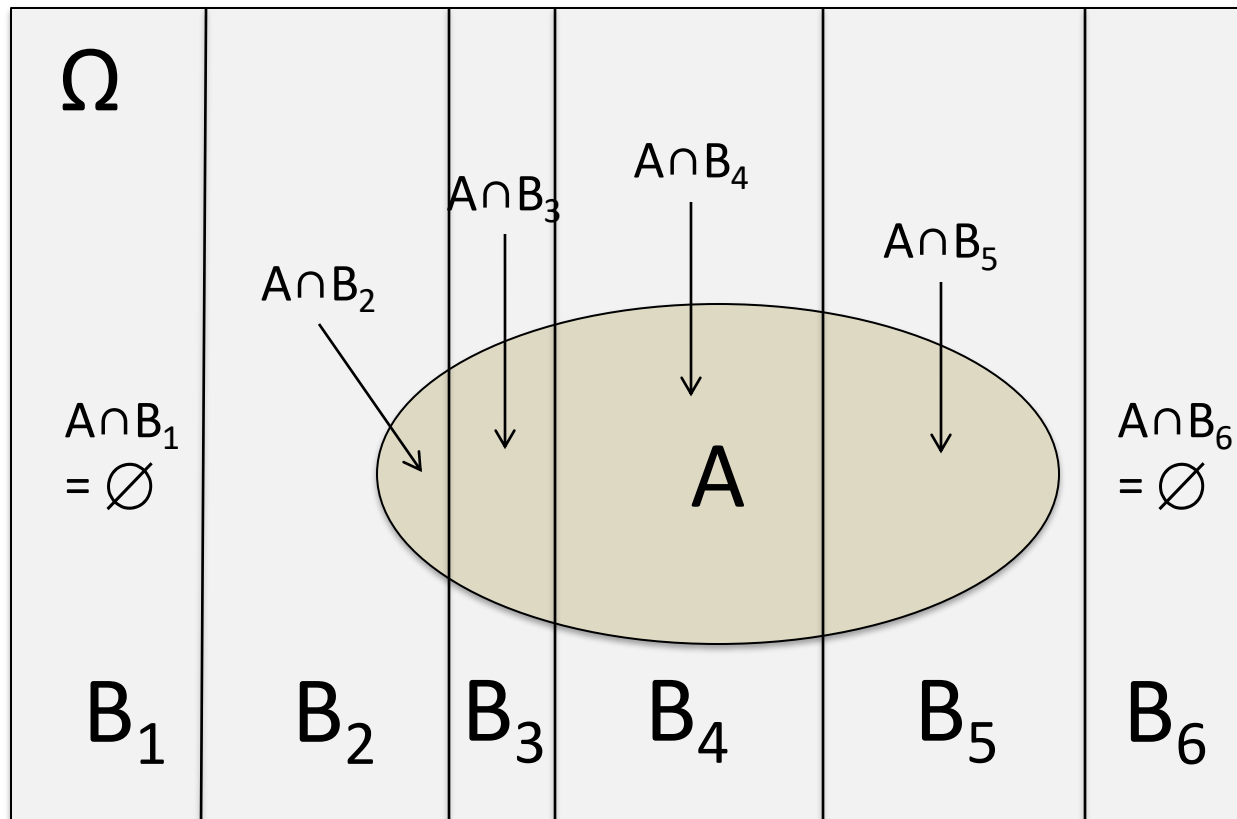


Law of Total Probability

- For A:

$$P(A) = P(A \cap B_1) + P(A \cap B_2) + \dots + P(A \cap B_n);$$

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots + P(A|B_n)P(B_n)$$

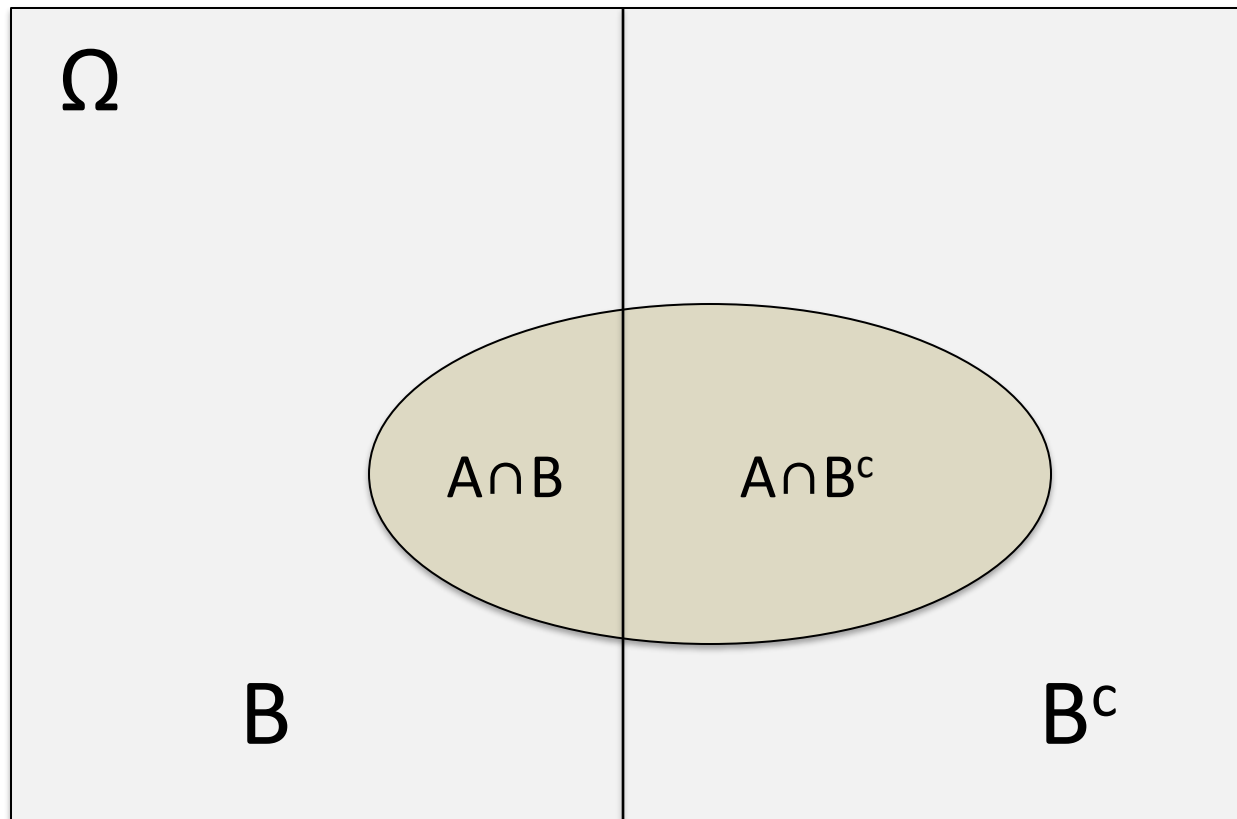


Law of Total Probability

- Special case:

$$P(A) = P(A \cap B) + P(A \cap B^c);$$

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c)$$



Law of Total Probability

- Example.** A die is rolled repeatedly to yield the number 1, and then a coin is tossed the same number of times. What is the probability that heads will not appear?

$A = \text{no heads}$

$B_1 = \text{roll die 1 time to get "1"}$

$B_2 = \text{roll die 2 times to get "1"}$

$\vdots$

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots$$

$$P(B_1) = \frac{1}{6} \quad P(B_3) = \left(\frac{5}{6}\right)^2 \times \frac{1}{6}$$

$$P(B_2) = \frac{5}{6} \times \frac{1}{6} \quad \vdots \quad P(B_k) = \left(\frac{5}{6}\right)^{k-1} \times \frac{1}{6}$$

$$P(A|B_k) = \left(\frac{1}{2}\right)^k$$

$$P(A) = \frac{1}{2} \times \frac{1}{6} + \left(\frac{1}{2}\right)^2 \times \frac{1}{6} \times \frac{1}{6} + \left(\frac{1}{2}\right)^3 \times \left(\frac{1}{6}\right)^2 \times \frac{1}{6} \\ + \dots$$

Law of Total Probability

- **Example.** A die is rolled repeatedly to yield the number 1, and then a coin is tossed the same number of times. What is the probability that heads will not appear?
- **Solution.** Let A be the event that heads will not appear, and B_i the event that the coin/die is tossed/rolled i times. Then $P(B_i) = (5/6)^{i-1}(1/6)$, $i=1,2,3,\dots$

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots$$

$$P(A|B_i) = (1/2)^i$$

$$\begin{aligned} P(A) &= (1/2)^1(1/6) + (1/2)^2(5/6)(1/6) + \dots + (1/2)^i(5/6)^{i-1}(1/6) + \dots \\ &= (1/12)[1/(1-5/12)] = 1/7 \end{aligned}$$